IF QCOH(X) IS EQUIVALENT TO A MODULE CATEGORY, X IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0GPE: sources/stacks-chow/tags.txt, source lines 20824–20824. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0B8J: sources/stacks-chow/tags.txt, source lines 14142–14142. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 005E: sources/stacks-chow/tags.txt, source lines 206–206. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01LA: sources/stacks-chow/tags.txt, source lines 2079–2079. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XF: sources/stacks-chow/tags.txt, source lines 2516–2516. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. If $QCoh(\mathcal{O}_X)$ is equivalent to the category of modules over a ring, then X is affine.*

Proof. Say $F : \mathrm{Mod}_R \to QCoh(\mathcal{O}_X)$ is an equivalence. Then $\mathcal{F} = F(R)$ has the following properties:

(1) it is a finitely presented $\mathcal{O}_X$-module (Lemma 0GPE),
(2) $\mathrm{Hom}_X(\mathcal{F}, -)$ is exact,
(3) $\mathrm{Hom}_X(\mathcal{F}, \mathcal{F})$ is a commutative ring,
(4) every object of $QCoh(\mathcal{O}_X)$ is a quotient of a direct sum of copies of $\mathcal{F}$.

Let $x \in X$ be a closed point. Consider the surjection

$$\mathcal{O}_X \to i_*\kappa(x)$$

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where the target is the pushforward of $\kappa(x)$ by the inclusion morphism $i : x \to X$. We have

$$\mathrm{Hom}_X(\mathcal{F}, i_*\kappa(x)) = \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \kappa(x))$$

This first by (4) implies that $\mathcal{F}_x$ is nonzero. From (2) we deduce that every map $\mathcal{F}_x \to \kappa(x)$ lifts to a map $\mathcal{F}_x \to \mathcal{O}_{X,x}$ (as it even lifts to a global map $\mathcal{F} \to \mathcal{O}_X$). Since $\mathcal{F}_x$ is a finite $\mathcal{O}_{X,x}$-module, this implies that $\mathcal{F}_x$ is a (nonzero) finite free $\mathcal{O}_{X,x}$-module. Then since $\mathcal{F}$ is of finite presentation, this implies that $\mathcal{F}$ is finite free of positive rank in an open neighbourhood of x (Modules, Lemma 0B8J). Since every closed subset of X contains a closed point (Topology, Lemma 005E) this implies that $\mathcal{F}$ is finite locally free of positive rank. Similarly, the map

$$\mathrm{Hom}_X(\mathcal{F}, \mathcal{F}) \to \mathrm{Hom}_X(\mathcal{F}, i_*i^*\mathcal{F}) = \mathrm{Hom}_{\kappa(x)}(\mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x, \mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x)$$

is surjective. By property (3) we conclude that the rank $\mathcal{F}_x$ must be 1. Hence $\mathcal{F}$ is an invertible $\mathcal{O}_X$-module. But then we conclude that the functor

$$\mathcal{H} \longmapsto \Gamma(X, \mathcal{H}) = \mathrm{Hom}_X(\mathcal{O}_X, \mathcal{H}) = \mathrm{Hom}_X(\mathcal{F}, \mathcal{H} \otimes_{\mathcal{O}_X} \mathcal{F})$$

on $QCoh(\mathcal{O}_X)$ is exact too. This implies that the first Ext group

$$\mathrm{Ext}^1_{QCoh(\mathcal{O}_X)}(\mathcal{O}_X, \mathcal{H}) = 0$$

computed in the abelian category $QCoh(\mathcal{O}_X)$ vanishes for all $\mathcal{H}$ in $QCoh(\mathcal{O}_X)$. However, since $QCoh(\mathcal{O}_X) \subset Mod(\mathcal{O}_X)$ is closed under extensions (Schemes, Section 01LA) we see that Ext^1 between quasi-coherent modules computed in $QCoh(\mathcal{O}_X)$ is the same as computed in $Mod(\mathcal{O}_X)$. Hence we conclude that

$$H^1(X, \mathcal{H}) = \mathrm{Ext}^1_{Mod(\mathcal{O}_X)}(\mathcal{O}_X, \mathcal{H}) = 0$$

for all $\mathcal{H}$ in $QCoh(\mathcal{O}_X)$. This implies that X is affine for example by Cohomology of Schemes, Lemma 01XF. □